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On the impact of adjusting the minimum life on receipt (MLOR) criterion in food supply chains☆

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ABSTRACT

The Minimum Life on Receipt (MLOR) is a widely used rule that imposes the minimum remaining age a food product must be delivered by the producer to the retailer. In practice, this rule is set by retailers and it is fixed, around 2/3 of the age of products regardless their shelf life. In this work, we study single and two echelon make-to-stock production-inventory problems for fixed-lifetime perishables. Mixed-integer linear optimization models are developed considering the MLOR rule both as decision variable and fixed parameter. When the MLOR rule is a variable, it is considered either a sole decision of the producer or a collaborative decision between retailer and producer. The goal of this work is to compare the supply chain performance considering this innovative setting of optimal MLOR (as a variable) against the traditional setting of fixed MLOR rule. The computational results suggest that allowing flexible MLOR rules according to the shelf life of products and the operational requirements of the producer benefit both entities in the supply chain. In particular, reducing the MLOR requirement in up to 12% does not interfere substantially with the average freshness of products arriving to the retailer, but reduces extensively surplus/waste generation at the producer while keeping a small amount of waste at the retailer.

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1. Introduction

About 15% of the total waste in the grocery retailing result from expired products [1]. Leading causes of such results include poor inventory management (e.g., ordering more than necessary due to uncertain demand), disregarding consumer behavior with respect to quality- and price-sensitivity, and the lack of coordination on supplier-retailer decisions due to, for example, the inexistence of information sharing [2–6].

To safeguard that, when reaching the shelf at stores, the products present sufficient available shelf life for consumers, retailers only accept products from suppliers if these are delivered within a pre-contracted minimum remaining shelf life. This requirement is known as the *Minimum Life On Receipt* (MLOR) and is set by retailers [7]. Products that are delivered to the retailer with a shelf life lower than MLOR are rejected and further removed from the food supply chain as food waste (or with a salvage value if they can be introduced in a secondary market). There is a general agreement of a 2/3 MLOR rule, i.e., only products with a remaining shelf of 2/3 are acceptable [8]. Consequently, this rule imposes that the upstream supply chain must produce and deliver the product within a period corresponding to 1/3 of its shelf life, directly influencing the production and replenishment decisions of suppliers.

The literature on food waste management in food supply chains has been increasing considerably, covering the identification of root causes and effects of food waste (e.g., de Moraes et al. [9]), the definition of best practices to prevent and reduce waste (e.g., Ribeiro et al. [10]) and the quantification of food waste and its impact on several performance indicators (e.g., Broekmeulen and van Donselaar [11]). Notwithstanding, works that evaluate the impact along the supply chain of the existing MLOR rule or that tackle alternative approaches for its definition are scarce in the literature. Furthermore, the MLOR rule strongly impacts the flexibility of production planning in practice but surprisingly no work has ever been devoted to study this issue. However, several food producers relate with the hard operational constraints imposed by the MLOR rule [7,12].

With the present work, we aim to provide a study tackling the MLOR rule with an in depth analysis of its impact in the performance of a supply chain. The running hypothesis of this paper is that by challenging the current practices for setting the MLOR, sub-

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stantial value can be unlocked in the food supply chain. The specific contributions to the literature are as follows:

1. Development of single- and two-echelon optimization models that address the MLOR rule as a decision variable;
2. Comparison of the new approach with the current practice of fixing the MLOR rule *a priori*, allowing to draw managerial insights that might help practitioners in establishing MLOR rules personalized to the type of food product;
3. Establishment of relationships between the optimized MLOR rule, product freshness and waste generation, considering different scenarios, namely shelf life, lot-size and consumer withdrawal.

With this work, we show strong evidence that relaxing the MLOR does not necessarily lead to a significant reduction of freshness. Actually, relaxing the MLOR rule by 10% produces an average freshness reduction of up to 4% for short shelf life products, whereas for longer shelf lives this reduction is under 1% on average. Also, the benefits of a flexible MLOR rule are more evident with increases in the lot-size and decreases in the shelf life, the average demand and the demand variability. In terms of food waste, the results show a clear potential of waste reduction by allowing flexible MLOR rules. In addition, waste reduction is more pronounced as the minimum lot-size decreases and the shelf life increases. An interesting insight of this work is that, while the producer always benefits from optimizing its decisions jointly with the retailer, it may sometimes lose with a relaxed MLOR. In opposition, the retailer always benefits from relaxing the MLOR rule; hence, a profit-sharing mechanism should be devised to induce an effective collaboration. We derive flexible MLOR rules using deterministic models, but evaluate their performance on a uncertain demand context.

The remainder of this paper is presented as follows. Section 2 presents a literature review of relevant works to contextualize the problem and the contributions of this work. The single- and two-echelons models are described and formulated in Section 3. The computational experiments and discussion of the results are provided in Section 4. Section 5 concludes with the main managerial insights of this work and directions for future research.

2. Related literature

This section reviews the literature that paves the ground for this work, focusing on papers that consider a MLOR rule (or an equivalent requirement) in the food supply chain. As mentioned above, the MLOR rule is an acceptance criterion defined by retailers concerning the minimum acceptable remaining shelf life a food product must be received. Therefore, MLOR is an age-dependent metric, which applies to products with a fixed shelf life.

So far, only the working paper of Mohamadi et al. [13] has considered the MLOR rule as a decision variable. The authors study a Stackelberg game between a retailer (leader) and a supplier (follower), investigating the effect of the MLOR agreement on the supply chain profits and waste. The results show that the supplier tends to send the freshest products to the retailer but both entities can benefit if the products sent are older. In such cases, the wholesale price of the supplier decreases but so it does the waste generated. Nevertheless, the authors only study this problem for a short shelf life product, examining a small change in the MLOR level.

The works of Akkas and Honhon [14] and Ketzenberg and Ferguson [2] study two-echelon problems between a supplier and a retailer, where the MLOR rule is set as a constraint. The work of Akkas and Honhon [14] focus on shipment policies from a warehouse to a retailer and their impact on individual and global profits, waste and freshness. The problem setting is a Vendor-Managed Inventory (VMI) formulated as an infinite horizon dynamic programming model maximizing profits. The authors analyse a warehouse which tends to ship the oldest products first to meet the demand of the retailer. The target products are consumer packaged goods (CPG) with a MLOR rule between 5 to 6 weeks. The warehouse is responsible for the waste generated at both echelons, reimbursing the retailer for expired products. The experiments conducted establish that the traditional policy – shipping oldest first (SOF) – is only optimal for waste minimization if the consumers at retailers select the products with a first-expired-first-out (FEFO) policy. However, if consumers follow a last-expired-first-out (LEFO) policy, shipping the oldest products first increase the odds of old units not being sold. Thus, the authors have proposed other shipment policies demonstrating when they outperform the SOF policy. Ketzenberg and Ferguson [2] study centralized and decentralized versions of a problem where a retailer shares the demand, inventory and ordering policy with a supplier. The goal is to evaluate the value of information (VOI) and the value of centralized control (VCC), which measures the marginal profits obtained with information sharing when compared with the opposite case – no information sharing. In the main setting, the retailer accepts products from the supplier with any age and the supplier receives the same unitary revenue independently of the product's age. This setting is latter extended to include a MLOR constraint and the supplier sensitivity to freshness such that it receives a higher revenue as the product shipped has larger remaining shelf-life. Moreover, the MLOR is a function of product's age and expected profit, which is dependent on freshness. The results show that increasing the price sensitivity to freshness reduces both the VOI and VCC.

An equivalent MLOR rule is the internal shelf life, which is the time interval a producer has available to produce and deliver a product to the retailer. In other words, is the difference between the shelf life of a product and the MLOR rule. This rule is studied in Pauls-Worm et al. [15] when solving a production problem with stochastic demand and a maximum internal shelf life constraint. To solve the production problem, the authors first derive approximate solutions using a deterministic mixed integer linear programming (MILP) and then assess their performance by Monte Carlo simulation. Among other parameters, three internal shelf lives are tested, namely 2, 3 and 4 periods. The results show that increasing the internal shelf life reduces the waste generated at the producer. A similar work is latter conducted in Pauls-Worm et al. [16].

Another equivalent MLOR rule is the minimum quality threshold. In opposition to the MLOR, which is dependent on age, the minimum quality rule is dependent on the product's quality, which varies with temporal and environmental factors. Thus, the minimum quality rules are mostly used for products with dynamic shelf life. For example, Rong et al. [17] develop a production and distribution problem of perishable products with quality decay. The quality decay is a function of temperature and time, and the problem is formulated as a MILP model minimizing total costs. The problem considers a set of producers, a set of distribution centres and a set of retailers. Each retailer defines its required minimum quality acceptance rule and products that do not meet this rule are discarded as waste. The results show that the higher is the quality requirement, the lower must be the shipment temperature. Leithner and Fikar [18] tackle the minimum quality threshold in a regional food supply chain consisting of producers and their cold stores, a distribution center (DC) and retailers. The authors develop a simulation-based decision support system (DSS) that integrates generic quality models and user-specified quality thresholds. At the cold stores and the DC, quality information is used to select the products that are shipped forward in the supply chain and the products that flow through alternative distribution channels, due to not complying with the quality requirement. At the retail-

ers, products that do not meet the quality threshold are counted as food waste. As the quality threshold increase at cold stores or at DC, the higher is the amount of products shifted for alternative channels and the lower is the waste at retailers. However, the food waste reduction at retailers is dependent on the withdrawal policy.

Some literature does not explicit any acceptance rule but rather investigates the impact of increasing the remaining shelf life of food products at retailers, or examine strategies to increase it. For example, Broekmeulen and van Donselaar [11] design mathematical expressions to quantify food waste, freshness and on shelf availability (OSA). Using empirical data from several retailers and from multiple product categories (fruit and vegetables, fresh meat and convenience products), the results show that increasing the store shelf life by one day potentiates a reduction of 43.1% in waste at retailer and an increase of 17% in freshness, or an increase of 3.4% in OSA. In another work, Kaipia et al. [19] show that changing from an order-based to a forecast-based replenishment method allowed to increase the remaining shelf life for milk from 2 to 3.4 days and for fish products from 11.1 to 14.7. Moreover, allowing an adequate information sharing between the supply chain, which reduces the non-value-adding operations, could increase the remaining shelf life of fresh poultry in 1 up to 3 days. Ferguson and Ketzenberg [1] consider that the remaining shelf life of products at a retailer is unknown upon receipt but if the supplier shares its inventory's age the retailer is able to improve its decision-making before placing an order. The results show that with information sharing the remaining shelf life upon receipt increased by 8%, leading to a reduction of 40% in the outdated products at the retailer.

A last research stream that motivates this work is found in practical reports that evidence that significant amounts of waste generated in food supply chains could be avoided with lesser strict MLOR rules. For example, a survey of international supply chains found that high amounts of fresh products are rejected at retailers by a slight delay of contracted delivery windows, of just minutes [20]. This evidences the dominant position of retailers in the food supply chain. Lee et al. [7] demonstrate the benefit of increasing the MLOR rule as a way to increase the remaining shelf life of products at retailers, but note that the MLOR criteria should be reviewed to reduce food waste at the upstream supply chain caused by products returns.

In light of this literature review, our work contributes significantly to the literature on food waste reduction in the context of sustainable food supply chains. The main novelty of our research relies on the optimization of the MLOR rule in production-inventory problems for different operational requirements (lot-size) and different product shelf lives, which contrasts with the above literature where the MLOR rule is set *a priori* as a parameter or examined for a limited range of shelf lives. Another distinction of this work is the analysis of food waste generation not only at one stage of the supply chain, but at both echelons - producer and retailer, and the investigation of the relationships between MLOR rule, freshness and food waste. It is also worth mentioning that while the main focus is on deterministic models, a rolling horizon approach is used to evaluate the effect of uncertain demand on the robustness of our results. Similar approaches can be found in relevant works in the field of perishables supply chains (e.g., Wang and Li [21], Adenso-D-az et al. [22]).

3. Mathematical formulations

In this section, two models are presented - a single-echelon and a two-echelon. The single-echelon model in Section 3.1 describes the problem of the producer who aims to define a production plan that maximizes profit while fulfilling the demand of a retailer. The two-echelon model in Section 3.2 describes an integrated producer-retailer problem, where the goal is to maximize the profits obtained by the production plan of the producer and the replenishment plan of the retailer. It is worth mentioning that we did not develop a single-echelon model for the retailer as it would require to assume the stock available at the producer and the respective ages. Instead, we describe the retailer's problem in the two-echelon model, thus avoiding unrealistic assumptions.

In practice, it is the retailer who decides the fixed MLOR value. A MLOR of 100% means that only products with age 0 are accepted by the retailer, whereas a MLOR of 0% means that products may be accepted with an age equal to their shelf life. The traditional retailer's strategy is to fix the MLOR rule around 66% [8], although higher values can be set [7]. In this work, we aim to compare such traditional strategy with two new settings where the MLOR is a decision variable. The first setting is an extreme one, where the MLOR rule is a decision of the producer. The second setting is an intermediate one, where the MLOR rule is a decision taken collaboratively between the retailer and the producer. The objective of testing different settings for deciding MLOR rules is to investigate what benefits they produce for the supply chain in comparison with the traditional setting. We expect to show that setting a MLOR rule dependent on the production requirements and product shelf life outperforms the traditional one-size-fits-all rule.

The mathematical notation used in the models is detailed in Table 1.

3.1. Problem of the producer

The problem of the producer consists on finding a production plan that maximizes profits for a given time horizon $T+1$ (i.e., the problem initializes at $t=0$). In addition, the problem aims to define the optimal value of the MLOR rule. The product ages along the time horizon, since its production ($a=0$) up to reaching its shelf life ($a=sl$). The producer plans when and how many units q_t to produce in order to attend the demand d_t of the retail channel in each period t. The minimum quantity to produce is set by the lot-size defined by the producer. The demand in each period t is satisfied with fractions $R_{a,t}$ of products with given age a. Products with age a that comply with the MLOR rule flow through the retail channel, which is given when variable $\theta_a = 1$. Surplus products flow through alternative channels if they are older or with equal age of the oldest products flowing to the retailer. The quantity available for alternative channels $A_{a,t}$ at each time period is given by the difference between the inventory $I_{a,t}$ and the total quantity delivered to the retailer. The lead-time is set at 0, meaning that products produced at period t flow through the retailer at that same period, becoming available to withdrawn at period $t+1$.

$$\max \sum_{t\in\mathcal{T}}\sum_{a\in\mathcal{A}}\Big[p\cdot R_{a,t}+s\cdot A_{a,t}-h\cdot(I_{a,t}-R_{a,t}-A_{a,t})\Big] - \sum_{t\in\mathcal{T}}\Big[u\cdot Y_t+c\cdot q_t\Big] \tag{1}$$

$$\sum_{a\in\mathcal{A}\setminus\{sl\}} R_{a,t} = d_t, \qquad \forall t\in\mathcal{T} \tag{2}$$

$$I_{a,t} = I_{a-1,t-1} - R_{a-1,t-1} - A_{a-1,t-1}, \qquad \forall a\in\mathcal{A}, t\in\mathcal{T}, a,t\geq 1 \tag{3}$$

$$I_{0,t} = q_t, \qquad \forall t\in\mathcal{T} \tag{4}$$

$$0 = I_{a,T} - R_{a,T} - A_{a,T}, \qquad \forall a\in\mathcal{A} \tag{5}$$

$$q_t \leq M\cdot Y_t, \qquad \forall t\in\mathcal{T} \tag{6}$$

Table 1
Mathematical notations.

Indexes and sets	
$t \in \mathcal{T}$	$\mathcal{T} = \{0, \ldots, T\}$
$a \in \mathcal{A}$	$\mathcal{A} = \{0, \ldots, sl\}, a \leq t$
Parameters	
Producer:	
p	price of product sold in retail channel
s	salvage value of product when shifting to alternative channels, $s < p$
u	setup cost
c	unitary production cost
h	unitary inventory holding cost
l	minimum lot-size (units)
Retailer:	
p'	price of product sold to consumers, $p' > p$
o	cost of waste disposal
v	variable ordering cost
f	fixed ordering cost
h'	cost of holding inventory
d_t	retailer demand or demand of the consumers in period t
ss	safety stock
α	fraction of LEFO consumers
sl	shelf life
T	time horizon
M	large number
Decision variables	
Producer:	
$I_{a,t}$	inventory of product with age a at the beginning of period t at the producer
$A_{a,t}$	amount of product with age a sold to alternative channels in period t
$R_{a,t}$	total quantity of product flowing through the retailer channel with age a in period t
q_t	quantity produced in period t
Y_t	1 if there is production in period t, and 0 otherwise
Retailer:	
Q_t	quantity ordered at the end of period t
$I'_{a,t}$	inventory of product with age a at the end of period t at the retailer
$W^L_{a,t}$	quantity of product with age a withdrawn by LEFO consumers in period t
$W^F_{a,t}$	quantity of product with age a withdrawn by FEFO consumers in period t
O_t	amount of products expired (with $a = sl$) at the end of period t
β_t	1 if an order is made at period t, and 0 otherwise
$\omega^L_{a,t}$	1 if inventory with age a is used to satisfy demand of LEFO consumers in period t
$\omega^F_{a,t}$	1 if inventory with age a is used to satisfy demand of FEFO consumers in period t
θ_a	1 if product with age a is accepted by the retailer, and 0 otherwise
$MLOR$	minimum life on receipt, in percentage, i.e. fraction of the total shelf life

$$q_t \geq l \cdot Y_t, \qquad \forall t \in \mathcal{T} \tag{7}$$

$$a \cdot \theta_a \leq (1 - MLOR) \cdot sl, \qquad \forall a \in \mathcal{A} \tag{8}$$

$$R_{a,t} \leq M \cdot \theta_a, \qquad \forall a \in \mathcal{A}, t \in \mathcal{T} \tag{9}$$

$$A_{r,t} \leq M(1 - \theta_a), \qquad \forall a, r \in \mathcal{A}, r < a, t \in \mathcal{T} \tag{10}$$

$$0 \leq MLOR \leq 1, \qquad \forall a \in \mathcal{A}, t \in \mathcal{T} \tag{11}$$

$$I_{a,t}, A_{a,t}, q_t, R_{a,t} \geq 0, \qquad \forall a \in \mathcal{A}, t \in \mathcal{T} \tag{12}$$

$$Y_t, \theta_a \in \{0, 1\}. \qquad \forall a \in \mathcal{A}, t \in \mathcal{T} \tag{13}$$

Objective function in (1) maximizes the total profits, given by the total revenues obtained with products flowing to retail and alternative channels minus the total costs of production (set up plus production costs) and inventory (holding costs). Constraints (2) ensure that all demand of the retail channel is met with products that have not reached their shelf life. Constraints (3) to (5) describe the inventory balance (quantity and age) throughout the time horizon. Constraints (6) ensures that setups occur whenever production occurs, and Constraints (7) establish the minimum quantity to produce (lot-size). Constraints (8) establish the relationship between variable θ and variable *MLOR*. Constraints (9) can be interpreted as follows. If $\theta = 1$, the product is fresh enough to be accepted by the retail channel; if $\theta = 0$, the age of products have surpassed the MLOR and thus they can no longer be assigned for the retail channel, only for the alternative channel. Constraints (10) ensure that products flowing to the alternative channel cannot be fresher than the oldest products flowing to the retailer channel. These constraints guarantee a certain robustness of the model as it ensures availability of products for the retailer complying with the MLOR rule in cases of uncertain demand. They also avoid unreasonable amounts of products to be sent to alternative channels. Constraints (11) to (13) define the domains of the decision variables.

3.2. Integrated producer-retailer problem

The integrated model combines the production planning of the producer with the replenishment planning of the retailer. The problem of the producer is the same as described above. The problem of the retailer consists on finding a replenishment plan that

maximizes the profits for a given time horizon $T+1$. The total quantity Q_t ordered in each period t depends on the inventory target level and the inventory available with different ages $I'_{a,t}$. Such quantity is satisfied by the producer with products of different ages accepted by the retailer $R_{a,t}$. The problem considers two types of consumers, namely LEFO and FEFO, and the parameter α determines the proportion of LEFO consumers. Accordingly, LEFO consumers select fresher products than FEFO consumers. The products that reach their shelf life sl at the end of period t cannot be sold in the next period. Thus, O_t represents the total quantity of outdated products.

$$\max \sum_{t\in\mathcal{T}}\sum_{a\in\mathcal{A}}\Big[p\cdot R_{a,t}+s\cdot A_{a,t}+p'\cdot(W^L_{a,t}+W^F_{a,t}) - h\cdot(I_{a,t}-R_{a,t}-A_{a,t})-h'\cdot I'_{a,t}\Big] - \sum_{t\in\mathcal{T}}\Big[u\cdot Y_t+c\cdot q_t+v\cdot Q_t+f\cdot\beta_t+o\cdot O_t\Big] \quad (14)$$

(2) − (13)

$$Q_t \geq ss+d_{t+1}-\sum_{a\in A\setminus\{sl\}} I'_{a,t}, \qquad \forall t\in\mathcal{T} \quad (15)$$

$$Q_t \leq M\cdot\beta_t, \qquad \forall t\in\mathcal{T} \quad (16)$$

$$\sum_{a\in A} W^L_{a,t} = d_t\cdot\alpha, \qquad \forall t\in\mathcal{T} \quad (17)$$

$$\sum_{a\in A} W^F_{a,t} = d_t\cdot(1-\alpha), \qquad \forall t\in\mathcal{T} \quad (18)$$

$$W^L_{a,t} \leq M\cdot\omega^L_{a,t}, \qquad \forall t\in\mathcal{T},\forall a\in\mathcal{A} \quad (19)$$

$$W^F_{a,t} \leq M\cdot\omega^F_{a,t}, \qquad \forall t\in\mathcal{T},\forall a\in\mathcal{A} \quad (20)$$

$$W^F_{b,t} \leq M\cdot(1-\omega^L_{a,t}), \qquad \forall t\in\mathcal{T},\forall a,b\in\mathcal{A}: b<a \quad (21)$$

$$I'_{a,t} = I'_{a-1,t-1}+R_{a-1,t-1}-W^L_{a,t}-W^F_{a,t}, \quad \forall t\in\mathcal{T}\setminus\{0\},\forall a\in\mathcal{A}\setminus\{0\} \quad (22)$$

$$M\cdot(1-\omega^L_{a,t}) \geq \sum_{\substack{b\in\mathcal{A}\\ b<a-1}}(I'_{b,t-1}+R_{b,t-1})-W^L_{a-1,t}, \forall t\in\mathcal{T}\setminus\{0\}, \quad \forall a\in\mathcal{A}, a>1 \quad (23)$$

$$M\cdot(1-\omega^F_{a,t}) \geq \sum_{\substack{b\in\mathcal{A}\\ b\geq a}}(I'_{b,t-1}+R_{b,t-1})-W^F_{a+1,t}, \forall t\in\mathcal{T}\setminus\{0\}, \quad \forall a\in\mathcal{A}, a<sl \quad (24)$$

$$Q_t = \sum_{a\in A} R_{a,t}, \qquad \forall t\in\mathcal{T} \quad (25)$$

$$\sum_{t\in\mathcal{T}} R_{a,t} \geq \theta_a, \qquad \forall a\in\mathcal{A} \quad (26)$$

$$O_t = I'_{sl,t}, \qquad \forall t\in\mathcal{T} \quad (27)$$

$$I'_{a,0} = 0, \qquad \forall a\in\mathcal{A} \quad (28)$$

$$I'_{0,t} = 0, \qquad \forall t\in\mathcal{T} \quad (29)$$

$$Q_t, O_t, I'_{a,t}, W^L_{a,t}, W^F_{a,t} \geq 0, \qquad t\in\mathcal{T},\forall a\in\mathcal{A} \quad (30)$$

$$\beta_t, \omega^L_{a,t}, \omega^F_{a,t} \in \{0,1\}. \qquad t\in\mathcal{T},\forall a\in\mathcal{A} \quad (31)$$

The objective function (14) maximizes the profit of both producer and retailer. The profits of the retailer are given by the difference between sales and total costs (inventory, ordering and outdated).

The integrated model incorporates all the constraints presented for the producer problem.

Constraints (15) to (31) represent the constraints of the retailer's problem. Constraints (15) determine the minimum amount of products ordered to the producer, which depends on the inventory level $I'_{a,t}$ and the target level given by $ss+d_{t+1}$. A periodic review of one period is considered and, therefore, the target level is defined as the safety stock plus the demand of the next period. Constraints (16) guarantee that an order is made in period t only if $\beta_t = 1$. Constraints (17) and (18) define the quantity withdrawn in each period by LEFO and FEFO consumers, respectively. Constraints (19) - (20) ensure that products of age a can only be withdrawn by LEFO and FEFO consumers if the corresponding variables $\omega^L_{a,t}$ and $\omega^F_{a,t}$ take value of 1, respectively. Constraints (21) reinforce that the freshest products available are firstly withdrawn by LEFO customers, although products with the same age can be withdrawn by both LEFO and FEFO consumers. Constraints (22) define the inventory balance within the time horizon. Constraints (23) ensure that products withdrawn by LEFO consumers are the freshest products available in a given period. In opposition, Constraints (24) ensure that products withdrawn by FEFO consumers are the oldest products available in a given period. Constraints (25) determine that the total quantity ordered in period t corresponds to the quantities of different ages a accepted by the retailer. Constraints (26) enforce that if a product with age a is accepted by the retailer, then a positive quantity $R_{a,t}$ effectively exists. Constraints (27) determine that the total quantity of outdated products correspond to the total quantity in inventory with age sl at the end of period t, which cannot be sold in the next period. Constraints (28) ensure that no inventory is available in period 0, while Constraints (29) ensure that products with age 0 cannot be part of the inventory of the retailer. The domain of the decision variables of the retailer are presented in Constraints (30)-(31).

4. Computational experiments

Five sets of computational experiments are conducted, each one led by a distinct motivation. The first set is presented in Section 4.1 and compares the supply chain performance obtained with the flexible and the fixed (traditional) MLOR rule. The second set is presented in Section 4.2 and compares the producer's performance in the single and two-echelon models. The third set presented in Section 4.3 investigates the individual benefits of producer and retailer and defines a simple profit sharing mechanism to fairly distribute such benefits, when necessary. The fourth set is presented in Section 4.4 and analyzes the effect of the consumer withdrawal policy. The last set of experiments is presented in Section 4.5 where a rolling horizon technique is applied to evaluate the robustness of our innovative policy in an uncertainty context. The performance indicators analyzed in the experiments are

the average freshness, the total amount of surplus products flowing to alternative channels, and the total profits. The average freshness is the normalized average remaining shelf life of the products flowing through the retailer, for the entire time horizon, calculated as $\frac{\sum_a \sum_t \frac{sl-a}{sl} R_{a,t}}{\sum_t d_t}$. Note that the total amount of surplus can be seen as wastage, in the sense that it does not reach the primary market, i.e., the retailer. In fact, while for some products secondary channels can be established, for others the only alternatives comprise donation or waste disposal.

To establish the relationships between MLOR rules and the different indicators, several instances are randomly generated for a time horizon (T) of 20 periods. The rationale to build these instances are as follows. To analyse the effect of lower and higher demands, three 5-instance groups are created. Group A considers an uniform distribution of [5, 20], group B a distribution of [10, 25] and group C a distribution of [10, 50]. Between groups B and C it is also possible to access the effect of low and high demand variability, respectively. Then, to analyse the effect of the producer's operational restrictions, increasing minimum lot-size values are set, namely {15, 45, 60} for group A and {25, 50, 75, 100} for groups B and C. Finally, to evaluate the effect of the type of product, three different shelf lives are set, namely {5, 10, 20}. Unless stated otherwise, the remaining instance parameters are kept constant in all tests as follows: $p = 4, s = 1, u = 5, c = 2, h = 0.1, p' = 5, o = 1, v = 4, f = 5, h' = 0.5, ss = 0, \alpha = 0.8, M = 200$.

In total, 165 instances are tested with both models. All models are coded in Python 3.6.3 and solved with Gurobi on a computer equipped with the processor Intel Core i7 and 16 GB of RAM.

4.1. Comparison of variable and fixed MLOR rules

The aim in this section is to evaluate the impact of deciding the optimal MLOR rule and compare it with the traditional setting of a fixed MLOR rule defined by the retailer. Thus, all instances are tested with both producer and integrated models presented in Sections 3.1 and 3.2, and run again with a fixed MLOR of 2/3 to mimic the traditional setting. All instances were run until optimality.

The results from the producer and integrated models are depicted in Tables 2 and 3, respectively. Both report the average indicators obtained for each 5-instances group, namely the objective function (OF), amount of product flowing to alternative channels (AC), average freshness and MLOR. They also show the difference (Δ) of each indicator obtained by fixing the MLOR rule in comparison with the variable MLOR, i.e., $\Delta = \frac{Indicator\ (Fixed\ MLOR) - Indicator\ (Optimal\ MLOR)}{Indicator\ (Fixed\ MLOR)}$. For values already in percentage, such as MLOR and Freshness, the difference is given by $\Delta = Indicator\ (Fixed\ MLOR) - Indicator\ (Optimal\ MLOR)$.

We anticipate that there are no outdated products at the retailer, and therefore we have not measured this indicator. This is a common assumption in papers dealing with production problems under deterministic demand (e.g., Amorim et al. [23]). This is also an expected result because the surplus product at the producer has a salvage value while at the retailer it only carries a cost. This assumption is in agreement with the works of Akkas et al. [4] and Leithner and Fikar [18]. We also anticipate that a fixed MLOR rule of 2/3 is hardly translated in 66% (of age), as the percentage is given considering a rounded up value of age. For example, for a shelf life of 5, the 2/3 MLOR (or 66%) corresponds to a remaining shelf life 3.3 periods. Rouding up this value, the MLOR is given by 4 periods of remaining shelf life or 80%. Accordingly, for a shelf life of 10, the fixed MLOR corresponds to 7 periods or 70%, and for a shelf life of 20, the fixed MLOR corresponds to 14 periods or 70%. Moreover, it is important to note that the fixed MLOR rule is set as a minimum value.

The results show that the MLOR increases with the shelf life and decreases with the lot-size for all groups of instances and for both producer and integrated problem. A shelf life of 5 periods seems to be the most critical as it always lead to an optimal MLOR lower than the traditional 2/3 rule, even when the lot-size is minimal. For the largest lot-size, the MLOR is around 20% for all instances groups, which means that a product is accepted by the retailer with a 1 period remaining shelf life. Although it may be unrealistic in practice, it is important to note that the models in Section 3 present a portion of FEFO consumers which would buy products near expiration. In Section 4.4 we further explore the influence of FEFO-LEFO ratios.

An important observation of the results is the unexpected relationship between MLOR and freshness. It would be expected to observe a relatively proportional increase of average freshness with MLOR. However, the results regarding the delta values for these two indicators show the following. First, when the MLOR increases by up to 12% in comparison with the optimal MLOR, no significant increase of freshness is observed. Even in some cases where the MLOR is forced to increase by 16%, the average freshness remains almost unchanged (e.g., Table 2c, 5 shelf life and 25 lot-size and Table 3c, 10 shelf life and 75 lot-size). Second, in the worst case, where the MLOR is forced to increase by 60%, the average freshness increases by around 30% at most. Note that when such cases occur, the largest amounts of products flowing to alternative channels also occurs, along with the highest reduction of profits. Finally, forcing MLOR increasing up to 24% leads to an increase of average freshness of up to 5%, while forcing up to 36% leads to a higher average freshness but no more than 10%.

Another evidence from the results is that the amount of products flowing to alternative channels tends to increase with shelf life and with lot-size. Again, a shelf life of 5 periods seems to be more critical than the others; in some instances, this is the only case when products flow to alternative channels (e.g., Tables 2b and 3 b). As the salvage value obtained from selling products in alternative channels is lower than the producing costs, it is expected to observe a decrease in the total profits as the amount of products flowing to alternative channels increase.

4.2. Comparison of single-echelon and two-echelon models

The aim in this section is to compare the producer's performance when the problem is modelled as a single-echelon and a two-echelon setting. We expect the producer to benefit from taking advantage on the knowledge of the retailers decisions instead of only acknowledge its daily demand. The differences between the indicators obtained with the integrated and producer models are reported in Table 4.

As expected, the profits for the producer increase if the problem considers the integrated decisions of producer and retailer, although this change is very smooth. Considering the single-echelon problem, the producer is obliged to deliver daily orders to the retailer to fulfil the daily demand. However, in the two-echelon setting, the producer only needs to deliver at some ordering moments, which are decided by the retailer. Thus, the producer may send more products to the retailer than the ones required to meet the demand of a single period. This allows the producer to reduce its inventory holding costs, which translates in higher profits. Although we have not addressed the single-echelon problem of the retailer, and thus, we cannot perform the same comparison as for the producer's performance, we argue that the retailer would not be in disadvantage. The reason is that the retailer might reduce the fixed costs associated with the ordering process, as it is no more required to order daily (an indirect assumption considered in the single-echelon setting of the producer).

Table 2
Indicators obtained with variable MLOR rule in the producer model and respective comparison with a fixed rule.

Lot-Size	15			45			60		
Shelf Life	5	10	20	5	10	20	5	10	20
OF	463.1	463.1	463.1	457.9	458.1	458.1	441.5	448.8	448.8
AC	0.0	0.0	0.0	0.2	0.0	0.0	6.2	0.8	0.8
Freshness	83.1%	91.6%	95.8%	72.3%	86.1%	93.1%	61.6%	80.7%	90.4%
MLOR	44.0%	72.0%	85.0%	24.0%	60.0%	77.0%	20.0%	52.0%	76.0%
Δ OF (%)	-1.3	0.0	0.0	-87.5	-1.2	0.0	-10.4	-604.5	0.0
Δ AC (%)	20.0	0.0	0.0	99.9	80.0	0.0	50.3	98.2	0.0
Δ Freshness (p.p.)	7.7	0.0	0.0	18.5	0.3	0.0	29.2	4.8	0.0
Δ MLOR (p.p.)	36.0	0.0	0.0	56.0	10.0	0.0	60.0	18.0	0.0

(a) Average results for group A instances

Lot-Size	25			50			75			100		
Shelf Life	5	10	20	5	10	20	5	10	20	5	10	20
OF	646.9	646.9	646.9	642.2	642.2	642.2	621.3	624.1	624.1	554.9	600.7	600.7
AC	0.0	0.0	0.0	0.0	0.0	0.0	1.2	0.0	0.0	46.6	0.0	0.0
Freshness	87.5%	93.8%	96.9%	78.1%	89.1%	94.5%	61.7%	81.1%	90.6%	59.7%	72.9%	86.5%
MLOR	60.0%	74.0%	85.0%	40.0%	68.0%	81.0%	20.0%	50.0%	76.0%	20.0%	32.0%	67.0%
Δ OF (%)	-0.5	0.0	0.0	-32.5	0.0	0.0	-201.3	-4.8	0.0	-17.2	-34.8	-0.3
Δ AC (%)	40.0	0.0	0.0	100.0	0.0	0.0	99.7	100.0	0.0	92.8	100.0	0.0
Δ Freshness (p.p.)	3.1	0.0	0.0	12.2	-0.1	0.0	28.7	4.3	0.0	30.7	12.5	-0.3
Δ MLOR (p.p.)	20.0	6.0	0.0	40.0	4.0	0.0	60.0	20.0	0.0	60.0	38.0	3.0

(b) Average results for group B instances

Lot-Size	25			50			75			100		
Shelf Life	5	10	20	5	10	20	5	10	20	5	10	20
OF	1187.0	1187.0	1187.0	1183.5	1183.5	1183.5	1171.7	1171.7	1171.7	1149.9	1150.5	1150.3
AC	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Freshness	92.5%	96.1%	98.0%	88.4%	94.2%	97.1%	81.9%	90.9%	95.5%	72.3%	86.1%	89.1%
MLOR	64.0%	82.0%	86.0%	56.0%	78.0%	86.0%	40.0%	66.0%	83.0%	20.0%	56.0%	67.0%
Δ OF (%)	-0.1	0.0	0.0	-2.8	0.0	0.0	-14.8	0.0	0.0	-48.2	-0.8	0.0
Δ AC (%)	20.0	0.0	0.0	100.0	0.0	0.0	100.0	0.0	0.0	100.0	60.0	0.0
Δ Freshness (p.p.)	0.3	0.0	0.0	3.0	0.0	0.0	8.3	-0.1	0.0	17.9	1.5	4.0
Δ MLOR (p.p.)	16.0	0.0	0.0	24.0	0.0	0.0	40.0	4.0	0.0	60.0	14.0	11.0

(c) Average results for group C instances

Table 3
Indicators obtained with variable MLOR rule in the integrated model and respective comparison with a fixed rule.

Lot-Size	15			45			60		
Shelf Life	5	10	20	5	10	20	5	10	20
OF	633.7	633.7	633.7	627.6	628.0	628.0	610.8	618.6	618.6
AC	0.0	0.0	0.0	1.1	0.6	0.6	6.2	1.1	1.1
Freshness	86.5%	93.2%	96.6%	77.2%	88.4%	94.2%	65.6%	82.9%	91.4%
MLOR	56.0%	78.0%	89.0%	28.0%	58.0%	77.0%	20.0%	46.0%	73.0%
Δ OF (%)	-0.3	0.0	0.0	-4.2	-0.1	0.0	-10.4	-1.6	0.0
Δ AC (%)	0.0	0.0	0.0	48.5	40.0	0.0	50.3	53.1	20.0
Δ Freshness (p.p.)	5.3	0.0	0.0	8.9	0.2	0.0	19.5	1.7	0.0
Δ MLOR (p.p.)	24.0	0.0	0.0	52.0	12.0	0.0	60.0	24.0	2.0

(a) Average results for group A instances

Lot-Size	25			50			75			100		
Shelf Life	5	10	20	5	10	20	5	10	20	5	10	20
OF	904.6	904.6	904.6	899.2	899.3	899.3	877.6	881.6	881.6	812.4	858.4	858.4
AC	0.0	0.0	0.0	0.2	0.0	0.0	2.2	0.0	0.0	46.6	0.0	0.0
Freshness	89.7%	94.9%	97.4%	79.7%	89.8%	94.9%	63.6%	82.1%	91.1%	61.4%	74.0%	87.0%
MLOR	64.0%	82.0%	91.0%	32.0%	58.0%	78.0%	20.0%	46.0%	74.0%	20.0%	38.0%	69.0%
Δ OF (%)	-0.1	0.0	0.0	-2.5	0.0	0.0	-8.5	-1.3	0.0	-14.1	-5.9	0.0
Δ AC (%)	0.0	0.0	0.0	53.3	40.0	0.0	89.1	40.0	0.0	0.0	80.0	0.0
Δ Freshness (p.p.)	1.1	0.0	0.0	6.9	0.1	0.0	22.1	1.5	0.0	22.5	8.2	0.0
Δ MLOR (p.p.)	16.0	0.0	0.0	48.0	12.0	0.0	60.0	24.0	2.0	60.0	32.0	1.0

(b) Average results for group B instances

Lot-Size	25			50			75			100		
Shelf Life	5	10	20	5	10	20	5	10	20	5	10	20
OF	1720.2	1720.2	1720.2	1716.5	1716.5	1716.5	1705.2	1705.5	1705.5	1682.3	1683.5	1683.5
AC	0.0	0.0	0.0	0.2	0.2	0.2	0.6	0.2	0.2	1.9	0.7	0.7
Freshness	92.6%	96.5%	98.2%	88.9%	94.5%	97.2%	82.6%	91.2%	95.6%	73.4%	86.6%	93.3%
MLOR	72.0%	86.0%	93.0%	64.0%	82.0%	90.0%	40.0%	58.0%	79.0%	28.0%	56.0%	78.0%
Δ OF (%)	0.0	0.0	0.0	-0.5	0.0	0.0	-2.5	0.0	0.0	-5.1	-0.3	0.0
Δ AC (%)	20.0	0.0	0.0	53.3	0.0	0.0	76.0	40.0	0.0	90.5	41.1	0.0
Δ Freshness (p.p.)	0.1	0.0	0.0	1.1	0.0	0.0	5.6	0.0	0.0	13.3	0.3	0.0
Δ MLOR (p.p.)	8.0	0.0	0.0	16.0	0.0	0.0	40.0	16.0	0.0	52.0	14.0	0.0

(c) Average results for group C instances

Table 4
Comparison of indicators obtained with the single- and two-echelon models.

Lot-Size	15			45			60		
Shelf Life	5	10	20	5	10	20	5	10	20
Δ OF	5.3	5.3	5.3	5.0	5.0	5.0	5.2	5.2	5.2
Δ AC	0.0	0.0	0.0	0.9	0.6	0.6	0.0	0.3	0.3
Δ Freshness	3.3%	1.7%	0.8%	4.9%	2.3%	1.1%	4.0%	2.2%	1.1%
Δ MLOR	12.0%	6.0%	4.0%	4.0%	-2.0%	0.0%	0.0%	-6.0%	-3.0%

(a) Average results for group A instances

Lot-Size	25			50			75			100		
Shelf Life	5	10	20	5	10	20	5	10	20	5	10	20
Δ OF	3.8	3.8	3.8	2.6	2.7	2.7	2.0	3.6	3.6	3.1	3.8	3.8
Δ AC	0.0	0.0	0.0	0.2	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0
Δ Freshness	2.2%	1.1%	0.5%	1.6%	0.8%	0.4%	1.9%	1.0%	0.5%	1.8%	1.1%	0.5%
Δ MLOR	4.0%	8.0%	6.0%	-8.0%	-10.0%	-3.0%	0.0%	-4.0%	-2.0%	0.0%	6.0%	2.0%

(b) Average results for group B instances

Lot-Size	25			50			75			100		
Shelf Life	5	10	20	5	10	20	5	10	20	5	10	20
Δ OF	1.4	1.4	1.4	1.3	1.3	1.3	1.6	1.9	1.9	0.5	1.1	1.4
Δ AC	0.0	0.0	0.0	0.2	0.2	0.2	0.6	0.2	0.2	1.9	0.7	0.7
Δ Freshness	0.2%	0.4%	0.1%	0.5%	0.2%	0.1%	0.7%	0.3%	0.2%	1.1%	0.5%	4.3%
Δ MLOR	8.0%	4.0%	7.0%	8.0%	4.0%	4.0%	0.0%	-8.0%	-4.0%	8.0%	0.0%	11.0%

(c) Average results for group C instances

Table 5
Individual gains of producer and retailer.

Lot-Size	15			45			60		
Shelf Life	5	10	20	5	10	20	5	10	20
Δ $OF(R)$	1.6	0.0	0.0	31.0	0.9	0.0	58.5	11.5	0.0
Δ $OF(P)$	0.2	0.0	0.0	-5.7	0.0	0.0	-1.5	-2.0	0.1
Δ $OF(P_0)$	5.3	5.3	5.3	5.0	5.0	5.0	5.2	5.2	5.2

Average results for group A instances

Lot-Size	25			50			75			100		
Shelf Life	5	10	20	5	10	20	5	10	20	5	10	20
Δ $OF(R)$	0.5	0.0	0.0	32.6	0.0	0.0	76.3	15.2	0.0	154.2	27.7	0.0
Δ $OF(P)$	0.0	0.0	0.0	-11.1	0.1	0.0	-8.4	-4.0	0.1	-54.0	19.4	0.0
Δ $OF(P_0)$	3.8	3.8	3.8	2.6	2.7	2.7	2.0	3.6	3.6	3.1	3.8	3.8

Average results for group B instances

Lot-Size	25			50			75			100		
Shelf Life	5	10	20	5	10	20	5	10	20	5	10	20
Δ $OF(R)$	0.0	0.0	0.0	6.0	0.0	0.0	38.0	0.0	0.0	91.2	4.8	0.0
Δ $OF(P)$	0.4	0.0	0.0	1.5	0.0	0.0	2.4	0.4	0.0	-10.5	-0.2	0.0
Δ $OF(P_0)$	1.4	1.4	1.4	1.3	1.3	1.3	1.6	1.9	1.9	0.5	1.1	1.4

Average results for group C instances

The results also show that the amount of products flowing to alternative channels is higher, or at least, the same in the two-echelon setting than in the single-echelon. As it happens with the profits, this change is smooth. Nevertheless, increasing the amount of products flowing to alternative channels is compensated by the reduction in inventory holding costs. There seems to be no relation with the increase in the amount of these products and the MLOR.

Another interesting observation is the relationship between the two-echelon setting and the average freshness. Note that within a two-echelon setting, the MLOR can be either higher, lower or the same as in the single-echelon, but the average freshness is always slightly higher. This can be explained by the increased possibility of the producer to send products to the retailer with different ages in a single order. Recall that the retailer considers both LEFO and FEFO consumers, and thus it has always demand for both young and old products. Also, LEFO consumers have higher demand than FEFO consumers, given by the parameter α. These assumptions are not considered in the single-echelon setting, where the producer is only guided by a FIFO issuing policy, shipping the oldest products first. Thus, mixing products with different ages in the same retailer's order is the reason why products reaching the retailer present a higher average freshness.

It is important to note that, although this section compares single- and two-echelon models, the objective function represents either the goal of the producer or the integrated goals of retailer and producer. Hence, no analysis is made to the individual benefits that both entities - retailer and producer - achieve in a collaborative setting. The next section is devoted to such analysis.

4.3. Comparison of producer and retailer benefits

In the previous sections, the objective function values shown for the two-echelon model represent the aggregated producer-retailer solutions. In particular, Table 3 reveals that an integrated solution is always better in a flexible MLOR setting than in the traditional fixed MLOR, whereas Table 4 shows that the producer

Table 6
Updated shares of retailer and producer gains.

Lot-Size	15			45			60					
Shelf Life	5	10	20	5	10	20	5	10	20			
R'	1.6	0.0	0.0	**30.3**	0.9	0.0	58.5	11.5	0.0			
P'	0.2	0.0	0.0	**0.7**	0.0	0.0	-1.5	-2.0	0.1			
Average results for group A instances												
Lot-Size	25			50			75			100		
Shelf Life	5	10	20	5	10	20	5	10	20	5	10	20
R'	0.5	0.0	0.0	**24.1**	0.0	0.0	**69.9**	**14.8**	0.0	**103.3**	27.7	0.0
P'	0.0	0.0	0.0	**8.5**	0.1	0.0	**6.4**	**0.4**	0.1	**50.9**	19.4	0.0
Average results for group B instances												
Lot-Size	25			50			75			100		
Shelf Life	5	10	20	5	10	20	5	10	20	5	10	20
R'	0.0	0.0	0.0	6.0	0.0	0.0	38.0	0.0	0.0	**81.3**	4.8	0.0
P'	0.4	0.0	0.0	1.5	0.0	0.0	2.4	0.4	0.0	**10.0**	-0.2	0.0
Average results for group C instances												

always benefits by integrating its problem with the retailer instead of optimizing its problem alone. However, the following question still remains - are both retailer and producer benefiting from a flexible MLOR rule? This section aims to answer this question by analyzing in detail the individual gains of retailer and producer. Table 5 reports the individual gains of retailer ($\Delta\ OF(R)$) and producer ($\Delta\ OF(P)$) achieved with a flexible MLOR in comparison with the fixed MLOR. The gains of producer ($\Delta\ OF(P_0)$) achieved by integrating its problem with the retailer are shown as well.

From the results in Table 5, it is possible to argue that the retailer always benefits from a flexible MLOR rule. Such benefit is due to lowering the holding costs in comparison with the setting of fixed MLOR rule. However, the same is not observed for the producer. While in some cases the producer also gains with a flexible MLOR (e.g., Table 5a, lot-size 15, shelf life 5), in others the producer incurs in losses (e.g., Table 5a, lot-size 45, shelf life 5). Such losses are due to an increase in the holding costs of products, due to Constraints (10) enforcing to keep in stock all products whose ages comply with the MLOR rule. In brief, the lower is the MLOR, the older the products can flow to the retailer and the larger the amount of products that can be kept in stock by the producer before flowing to alternative channels, leading to higher holding costs. Note that, in the fixed MLOR setting, the products flowing to the retailer are fresher than in the flexible MLOR setting. Thus, in the latter case, the products remain in stock for a longer period until being rejected by the retailer, and only then they can flow to alternative channels. Hence the higher holding costs of products flowing to alternative channels in the flexible MLOR setting. More importantly, in such cases, the producer's losses surpass the producer's gains achieved by integrating its problem with the retailer.

Based on the reported results, it is possible to argue that while the producer always benefits from integrating its problem with the retailer, it may not always benefit from changing the MLOR rule. However, it is well-known that collaboration must induce gains for all enrolled participants. Therefore, to induce collaboration, some incentives must be offered to participants that, otherwise, loose. In this section, we propose a simple strategy of profit-sharing such that all participants - retailer and producer, benefit from collaboration in a flexible MLOR setting. This strategy is depicted in Fig. 1 and works as follows. If the producer loses with the integrated problem with flexible MLOR (i.e. $\Delta\ OF(P) < 0$) these losses are compared against their gains achieved by integrating the problem with the retailer ($\Delta\ OF(P_0)$). Afterwards, only if these losses are higher than the gains, the shares of each participant are updated. In brief, the retailer offers to the producer an incentive that corresponds to the difference between the losses and gains of the producer.

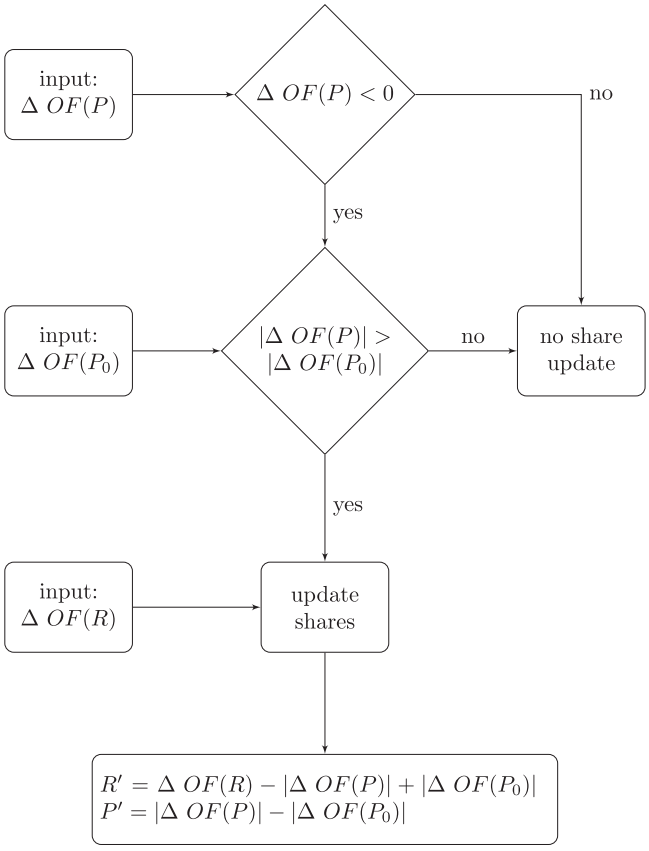


Fig. 1. Profit-sharing strategy.

Table 6 reports the updated shares of both retailer (R') and producer (P') after applying the profit-sharing strategy presented in Fig. 1, with the bold values presenting the shares that were adjusted accordingly. The results show that few shares adjustments are necessary. Even when the producer looses in an integrated problem with flexible MLOR, the gains initially obtained by integrating its problem with the retailer are higher; thus no share updated is required (e.g., Table 6a, lot-size 60, shelf lives 5 and 10). Furthermore, the few cases when shares must be updated com-

Table 7
Comparison of indicators obtained with $\alpha = 0.8$ and $\alpha = 1.0$.

Lot-Size	15			45			60		
Shelf Life	5	10	20	5	10	20	5	10	20
Δ OF	0	0	0	1.244	1.18	1.18	0	0.74	0.74
Δ AC	0	0	0	-0.68	-1.4	-1.4	0	-0.72	-0.72
Δ Freshness	0.0%	0.0%	0.0%	1.0%	-0.3%	-0.1%	0.0%	-0.2%	-0.1%
Δ MLOR	0.0%	0.0%	0.0%	-8.0%	-4.0%	-4.0%	0.0%	0.0%	0.0%

(a) Average difference between group A instances

Lot-Size	25			50			75			100		
Shelf Life	5	10	20	5	10	20	5	10	20	5	10	20
Δ OF	0	0	0	0.592	0.684	0.684	0.744	2.092	2.092	0	0	0
Δ AC	0	0	0	-0.6	-0.8	-0.8	0.6	-1	-1	0	0	0
Δ Freshness	0.0%	0.0%	0.0%	0.4%	0.2%	0.1%	0.9%	0.1%	0.0%	0.0%	0.0%	0.0%
Δ MLOR	0.0%	0.0%	0.0%	-12.0%	-12.0%	-8.0%	0.0%	-4.0%	-1.0%	0.0%	0.0%	0.0%

(b) Average difference for group B instances

Lot-Size	25			50			75			100		
Shelf Life	5	10	20	5	10	20	5	10	20	5	10	20
Δ OF	0.444	0.444	0.444	0.16	0.16	0.164	1.408	1.684	1.684	2.936	2.46	2.46
Δ AC	-0.6	-0.6	-0.6	0.04	0.04	0.04	0	-0.4	-0.4	1.72	0.72	0.72
Δ Freshness	-0.1%	0.1%	-0.1%	0.1%	0.0%	0.0%	0.5%	0.2%	0.1%	2.3%	0.7%	0.3%
Δ MLOR	-8.0%	-4.0%	-2.0%	0.0%	0.0%	-1.0%	-4.0%	-14.0%	-7.0%	0.0%	8.0%	4.0%

(c)Average difference for group C instances

Table 8
Comparison of indicators obtained with $\alpha = 0.8$ and $\alpha = 0.5$.

Lot-Size	15			45			60		
Shelf Life	5	10	20	5	10	20	5	10	20
Δ OF	0	0	0	0.614	0.91	0.91	0	0.36	0.36
Δ AC	0	0	0	-0.18	-0.7	-0.7	0	0.08	0.08
Δ Freshness	0.0%	0.0%	0.0%	0.3%	0.0%	0.0%	0.0%	0.1%	0.1%
Δ MLOR	0.0%	0.0%	0.0%	-4.0%	-6.0%	-5.0%	0.0%	12.0%	7.0%

(a) Average difference between group A instances

Lot-Size	25			50			75			100		
Shelf Life	5	10	20	5	10	20	5	10	20	5	10	20
Δ OF	0	0	0	0.492	0.584	0.584	-1.056	0.222	0.222	0	0	0
Δ AC	0	0	0	-0.6	-0.8	-0.8	0.6	-0.5	-0.5	0	0	0
Δ Freshness	0.0%	0.0%	0.0%	-0.2%	-0.1%	-0.1%	-0.1%	-0.2%	-0.1%	0.0%	0.0%	0.0%
Δ MLOR	0.0%	0.0%	0.0%	-8.0%	-10.0%	-7.0%	0.0%	-12.0%	-5.0%	0.0%	0.0%	0.0%

(b) Average difference for group B instances

Lot-Size	25			50			75			100		
Shelf Life	5	10	20	5	10	20	5	10	20	5	10	20
Δ OF	0.324	0.324	0.324	0.16	0.16	0.164	0.748	0.894	0.894	1.646	1.17	1.17
Δ AC	0	0	0	0.04	0.04	0.04	-0.2	-0.4	-0.4	0.12	0.72	0.72
Δ Freshness	-0.2%	0.1%	0.0%	0.1%	0.0%	0.0%	0.2%	0.1%	0.0%	0.9%	0.5%	0.3%
Δ MLOR	0.0%	0.0%	0.0%	0.0%	0.0%	-1.0%	4.0%	-4.0%	-2.0%	0.0%	12.0%	6.0%

(c) Average difference for group C instances

prise those where shelf life is 5, with the exception of one where shelf life is 10 (see Table 6b, lot-size 75). Nevertheless, the updated shares of the retailer are still very high in comparison with the shares of the producer, which indicate that the retailer would always be willing to collaborate.

4.4. Effect of consumer withdrawal policy

In the previous sections, we have considered a percentage of LEFO consumers of 80% ($\alpha = 0.8$). This default value was collected from Chung and Li [24] which state that about 88% of consumers check for the expiration dates of products on a store shelf. In this section, we compare this default value with $\alpha = 1.0$, assuming that all consumers follow a LEFO withdrawal policy, and with $\alpha = 0.5$, assuming that there is an equal percentage of FEFO and LEFO consumers. A setting with pure FEFO consumers is not investigated as it does not seem to be realistic for perishables products. Tables 7 and 8 report the difference in each indicator obtained with the initial LEFO percentage and with the new percentages of $\alpha = 1.0$ and $\alpha = 0.5$, respectively.

Overall, the results show that no significant differences are obtained in the three indicators - average freshness, surplus flowing for alternative channels and profits, which lead to argue that the ratio LEFO-FEFO consumers does not influence the performance of the supply chain.

In more detail, it is possible to observe in Table 7 that, in general, as expected, the MLOR tends to increase with an increase in the percentage of LEFO consumers, given by a negative value of Δ MLOR. However, the average freshness shows an erratic behaviour, as in some cases it reduces with increasing MLOR (e.g., Table 7a, lot-size 45, shelf life 5) and in others it increases (e.g., Table 7a, lot-size 45, shelf life 10). However, this change in the average freshness is very smooth, less than +/-1%. In opposition, the results in Table 8 evidence that it is not possible to establish any relationship between the increase in the percentage of FEFO consumers

Table 9
Average results of the rolling horizon approach with demand variation up to 10%.

		Model		Model		Δ(%)		Model		Δ(%)		Model		Δ(%)		Model		Δ(p.p.)	
sl	*l*	MLOR	sd	OF	sd	OF	sd	W-P	sd	W-P	sd	W-R	sd	W-R	sd	F	sd	F	sd
5	15	56%	9%	586.3	15.5	-1	3	2.4	1.4	-30	80	0.5	0.7	-6	30	83%	1%	4	2
	45	28%	11%	580.6	16.6	-4	2	5.2	2.6	23	40	0.9	1.0	3	45	75%	1%	9	1
	60	20%	0%	564.1	17.4	-11	3	10.0	2.3	41	15	1.7	1.2	-4	69	64%	1%	19	1
10	15	78%	4%	587.2	15.6	0	1	2.2	1.4	-40	74	0.5	0.6	0	1	90%	1%	0	0
	45	54%	9%	582.5	14.6	0	2	4.1	2.3	20	37	0.5	0.7	0	0	85%	2%	1	1
	60	46%	17%	580.7	13.4	-2	1	4.1	2.1	26	33	0.6	0.6	0	13	81%	1%	1	1
20	15	89%	2%	585.8	15.8	1	1	2.2	1.4	-79	143	0.6	0.7	-1	12	93%	1%	0	0
	45	77%	4%	583.4	16.5	0	1	3.9	2.4	-52	165	0.6	0.7	0	8	91%	1%	0	1
	60	73%	8%	578.2	14.7	0	1	4.7	2.0	-29	86	0.5	0.6	0	0	89%	1%	0	0

(a) For group A instances

		Model		Model		Δ(%)		Model		Δ(%)		Model		Δ(%)		Model		Δ(p.p.)	
sl	l	MLOR	sd	OF	sd	OF	sd	W-P	sd	W-P	sd	W-R	sd	W-R	sd	F	sd	F	sd
5	25	64%	9%	832.5	18.4	-3	20	3.5	1.8	-5	56	0.5	0.8	0	19	85%	2%	1	3
	50	32%	18%	832.6	21.3	-3	2	4.3	2.3	14	48	0.8	1.0	12	34	76%	1%	7	1
	75	20%	0%	811.2	22.1	-9	3	8.6	2.8	71	8	1.8	1.2	-19	80	62%	1%	21	1
	100	20%	0%	746.2	23.7	-15	3	54.4	3.6	-4	5	1.9	1.4	47	36	60%	1%	22	1
10	25	82%	4%	826.9	37.5	1	3	3.5	1.9	-59	128	0.6	0.8	-2	10	90%	2%	0	2
	50	56%	9%	835.4	20.5	0	1	3.8	2.2	21	29	0.6	0.9	-3	29	87%	1%	0	1
	75	48%	18%	821.8	18.6	-2	1	4.8	2.4	-7	104	0.7	0.8	10	34	80%	1%	2	1
	100	38%	4%	804.5	17.0	-7	2	4.9	2.7	61	17	0.5	0.7	19	31	72%	1%	8	1
20	25	91%	2%	832.3	22.1	0	1	3.2	1.9	-65	108	0.5	0.7	-6	33	94%	1%	0	1
	50	79%	7%	835.5	18.3	0	1	3.4	1.8	-26	63	0.5	0.7	1	4	92%	1%	0	0
	75	74%	9%	821.1	18.7	0	1	4.7	2.2	-32	71	0.5	0.7	1	7	88%	1%	0	1
	100	69%	2%	802.6	16.0	0	0	5.2	2.3	-16	24	0.6	0.8	0	0	85%	1%	0	0

(b) For group B instances

		Model		Model		Δ(%)		Model		Δ(%)		Model		Δ(%)		Model		Δ(p.p.)	
sl	l	MLOR	sd	OF	sd	OF	sd	W-P	sd	W-P	sd	W-R	sd	W-R	sd	F	sd	F	sd
5	25	72%	18%	1498.2	231.1	0	0	6.7	3.2	9	5	7.3	20.1	0	0	87%	5%	0	0
	50	64%	9%	1559.2	91.5	-9	55	7.0	3.4	12	42	2.1	7.0	-103	627	84%	2%	1	5
	75	40%	20%	1571.4	38.6	-4	2	11.8	5.4	34	36	1.0	1.2	6	27	79%	1%	5	1
	100	28%	11%	1556.2	34.9	-6	2	13.0	5.6	68	13	1.8	1.8	-2	51	70%	1%	13	1
10	25	86%	9%	1528.1	164.4	3	9	6.3	3.1	-122	243	2.7	9.0	-102	575	91%	4%	2	6
	50	82%	4%	1572.8	38.5	0	1	7.0	3.3	-66	173	0.9	1.1	0	2	90%	1%	0	1
	75	58%	18%	1572.8	39.8	0	1	10.9	5.3	5	40	1.0	1.3	-3	17	87%	1%	0	1
	100	52%	11%	1562.6	35.9	-1	2	9.5	5.7	26	55	0.8	1.1	0	0	83%	1%	1	1
20	25	93%	4%	1516.5	192.7	3	11	6.1	2.8	-205	444	2.5	7.8	-26	123	92%	5%	1	5
	50	91%	2%	1543.2	148.4	2	8	6.7	3.4	-166	389	2.1	5.9	-6	56	92%	4%	1	4
	75	79%	9%	1575.1	38.5	0	1	10.3	5.2	-56	94	0.8	1.1	-5	43	92%	1%	0	0
	100	76%	5%	1565.4	34.8	0	1	9.8	5.4	-39	91	1.0	1.2	1	11	90%	1%	0	0

(c) For group C instances

and MLOR, as the change in this value is very erratic, even for the same group of instances (e.g., Table 8(a) and c).

Another evidence from these results is that the profits obtained with 80%-20% LEFO-FEFO ratio are always equal or higher than those obtained with higher or lower LEFO percentages. Nevertheless, the change in profits is very small, in the order of less than 3 units, at most. Also, it is not possible to establish any relationship between the surplus amount and the percentage of LEFO consumers.

Despite the lack of relationships between the MLOR and the consumer withdrawal policy suggested by the results, the same trend established in the previous sections between MLOR and average freshness can be observed. In fact, it is possible to evidence that a change in the MLOR, either positive or negative, produces a much smaller change in the average freshness, always lower than 1%.

4.5. Rolling horizon

This section aims to evaluate the robustness of our innovative approach by comparing the deterministic results presented in the previous sections with the results provided by a new set of experiments considering uncertain demand. The experiments with uncertain demand use a rolling horizon approach and it is applied to a slightly different model than the one presented in Section 3.2, which is provided in Appendix A. This new model accounts for lost sales in the objective function and the original deterministic demand is substituted for a random demand d^*. Also, to prevent infeasible solutions, some equality constraints are changed to inequalities.

With the rolling horizon approach, the model is solved iteratively for each period while the decisions of the previous periods are fixed. The rolling horizon approach has been advocated as a user-friendly method widely applied in production and inventory management problems [25]. In particular for our problem, the model runs firstly with the deterministic demand for the entire time horizon, as in the previous experiments. We then fix the MLOR rule obtained as well as the decisions of period $t = 0$, namely the quantity to produce, the quantity ordered by the retailer and the quantities of products and their ages sent to the retailer. Then, we generate a random demand for period $t = 1$ keeping all the subsequent periods with the same demand as the deterministic instance. The model runs with this new demand pattern and the decisions of period $t = 1$ are then fixed. Then, this iterative process is repeated for all remaining periods until reach-

Table 10
Average results of the rolling horizon approach with demand variation up to 20% .

		Model		Model		Δ(%)		Model		Δ(%)		Model		Δ(%)		Model		Δ(p.p.)	
sl	l	MLOR	sd	OF	sd	OF	sd	W-P	sd	W-P	sd	W-R	sd	W-R	sd	F	sd	F	sd
5	15	56%	9%	558.0	21.6	2	15	3.5	2.0	-22	104	1.2	1.6	-3	35	81%	2%	3	8
	45	28%	11%	548.4	26.2	-4	3	8.4	3.9	18	44	1.8	1.8	-17	133	74%	2%	9	1
	60	20%	0%	531.8	23.4	-11	4	12.6	3.7	27	29	3.2	1.9	-8	97	64%	1%	19	3
10	15	78%	4%	553.7	21.0	0	1	3.8	2.1	-97	199	1.0	1.2	-1	18	88%	1%	0	1
	45	54%	9%	553.3	22.7	0	2	5.8	3.6	12	55	1.1	1.3	-3	22	83%	2%	1	2
	60	46%	17%	544.5	21.4	-2	3	6.7	2.9	29	32	1.2	1.3	2	19	79%	2%	2	1
20	15	89%	2%	554.5	20.9	1	2	3.8	1.9	-176	292	1.0	1.2	-2	45	92%	2%	0	1
	45	77%	4%	550.6	22.7	0	1	6.2	3.6	-42	108	1.1	1.4	2	14	90%	2%	0	1
	60	73%	8%	548.8	21.4	0	1	6.6	3.4	-15	60	1.0	1.3	0	4	88%	1%	0	1

(a) For group A instances

		Model		Model		Δ(%)		Model		Δ(%)		Model		Δ(%)		Model		Δ(p.p.)	
sl	l	MLOR	sd	OF	sd	OF	sd	W-P	sd	W-P	sd	W-R	sd	W-R	sd	F	sd	F	sd
5	25	64%	9%	794.8	28.5	0	13	5.5	2.8	-4	78	1.2	1.5	2	14	83%	2%	1	6
	50	32%	18%	785.9	33.5	-2	3	8.6	6.0	-37	242	1.8	1.6	6	40	74%	2%	8	2
	75	20%	0%	764.6	32.5	-8	4	14.9	5.7	58	12	3.2	2.4	-34	172	61%	1%	21	1
	100	20%	0%	697.6	31.6	-15	4	60.3	6.1	-12	9	3.9	2.6	48	32	60%	1%	23	1
10	25	82%	4%	774.9	84.3	2	10	6.0	2.9	-109	204	2.1	6.4	-1	9	88%	4%	0	4
	50	56%	9%	788.7	30.6	0	2	6.4	3.7	11	34	1.2	1.4	0	11	84%	2%	0	1
	75	48%	18%	781.9	25.3	-1	2	7.1	3.7	-2	117	1.1	1.3	9	34	78%	1%	2	1
	100	38%	4%	757.8	24.5	-6	3	8.9	4.4	54	34	1.1	1.3	18	29	70%	1%	9	1
20	25	91%	2%	787.0	67.8	1	8	5.5	2.4	-190	451	1.5	3.3	-194	1082	91%	4%	0	3
	50	79%	7%	790.9	29.0	0	2	6.3	3.5	-119	395	1.2	1.5	-5	40	89%	1%	0	1
	75	74%	9%	779.6	27.0	0	1	7.6	3.6	-36	106	1.0	1.3	1	7	86%	2%	0	1
	100	69%	2%	757.1	25.6	0	0	8.8	4.3	-22	32	1.0	1.3	0	0	83%	1%	0	0

(b) For group B instances

		Model		Model		Δ(%)		Model		Δ(%)		Model		Δ(%)		Model		Δ(p.p.)	
sl	l	MLOR	sd	OF	sd	OF	sd	W-P	sd	W-P	sd	W-R	sd	W-R	sd	F	sd	F	sd
5	25	72%	18%	1444.4	249.5	-1	4	10.9	5.6	8	6	7.3	22.5	-1	6	85%	5%	0	1
	50	64%	9%	1486.5	132.3	-1	2	10.3	4.9	12	54	3.9	11.3	-6	37	82%	3%	2	1
	75	40%	20%	1499.5	57.0	-4	3	17.0	8.1	22	65	2.3	2.7	-6	47	77%	1%	6	1
	100	28%	11%	1467.5	53.7	-6	3	22.3	8.3	51	23	3.6	3.2	37	156	68%	1%	14	3
10	25	86%	9%	1481.8	133.7	2	7	9.7	5.4	-155	219	2.8	6.6	26	153	89%	4%	0	9
	50	82%	4%	1494.1	54.5	0	2	11.3	5.9	-68	141	2.1	2.3	-1	12	88%	1%	0	4
	75	58%	18%	1497.4	54.2	0	2	14.7	7.8	8	33	1.8	2.4	-1	9	85%	2%	0	1
	100	52%	11%	1484.2	47.9	-1	2	15.9	7.1	29	22	1.7	2.3	0	2	81%	1%	1	1
20	25	76%	42%	1435.4	215.7	5	14	10.1	5.4	-392	832	3.5	6.8	35	181	89%	7%	3	10
	50	91%	2%	1498.9	100.1	1	5	10.8	5.5	-295	590	2.2	3.1	8	40	91%	3%	-1	6
	75	79%	9%	1498.4	49.5	0	1	15.3	6.8	-104	183	2.4	3.1	5	43	90%	2%	0	1
	100	76%	5%	1483.7	52.4	0	1	15.8	7.6	-43	118	1.8	2.3	-2	9	88%	1%	0	1

(c) For group C instances

ing period $t = T$. The random demand in each period is allowed to take any value within +/-10% or +/- 20% of the deterministic demand. Hence, the real (random) demand can be the same, lower or higher than the expected (deterministic) demand. As with the experiments with deterministic demand, and to simplify the experiments with the rolling horizon approach, we consider a safety stock of 0. This is a common assumption when testing rolling horizon procedures (e.g., Clark [26], Tresoldi and Ceselli [27]). This procedure runs 30 times for each instance and the average results are reported in Tables 9 to 10. The indicators presented refer to total profits (OF), total waste/surplus at the producer (W-P), total waste/surplus at the retailer (W-R) and average freshness (F). For each indicator, the average value and respective standard deviation obtained with the model are provided, as well as the percentage differences Δ from the values obtained with the fixed rule of MLOR = 66%.

Overall, these results show the same trends of the deterministic results presented in Table 3. When the optimal MLOR rule is lower than the standard practice of 66%, the average delta profits increases (given by a negative value), and such an increase is more pronounced with increasing MLOR flexibility (i.e., lowering MLOR value). On the other hand, when the optimal MLOR is more rigid than the standard practice (i.e., MLOR > 66%), the average delta profits are nearly zero or slight losses may occur, but not higher than 5% on average (e.g., Table 10c, shelf life 20, lot size 25).

Regarding the comparison of waste/surplus generated at the producer and retailer, we have seen with the deterministic results that this is null at the retailer, which is an expected result in a deterministic context. With the rolling horizon results, it is possible to observe a different picture, with waste/surplus generated at both entities. Still, in general, the average amount of waste/surplus at the retailer is smaller than the waste/surplus generated at the producer, which agrees with the deterministic results in Table 3. Another evidence from Tables 9 to 10 in accordance with the deterministic results is that, in general, the more flexible is the MLOR rule (the lower the MLOR value), the higher is the delta waste at the producer, which means a larger positive difference between the standard and the flexible systems.

Also concerning the average freshness, the results show a good agreement with the deterministic results, namely that increasing the MLOR rule flexibility has a disproportional effect on the average reduction. Although it can be observed larger percentage reductions on the freshness parameter when the MLOR rule is relatively reduced (e.g., see Table 9a, for a MLOR of 56%, i.e., a reduc-

tion of 10% in MLOR, the average freshness reduction is 4%), this only happens when the shelf life is 5. In fact, when the shelf life is 10 or 20, these MLOR reductions are around 1%, at most. Another interesting result is that reducing MLOR flexibility (i.e., imposing a MLOR higher than 66%) brings no advantage to the average freshness, as shown by the zero percentage difference obtained in most cases.

Based on the results from the rolling horizon approach, we can argue that our previous conclusions obtained with the deterministic approach, namely that a flexible MLOR rule is more beneficial for a producer-retailer system in terms of profits and waste than a standard fixed MLOR rule, are robust, since the same trends in the results with uncertain demand are observed.

5. Conclusions and future research

This paper explores the opportunity of deciding the optimal MLOR rule instead of considering this rule as a strict, standardized and fixed parameter imposed by retailers. The main hypothesis of this paper is that allowing the optimization of the MLOR rule considering different operational restrictions and products shelf lives benefits the food supply chain in terms of profits and waste/surplus generation. A single-echelon model representing the problem of a producer and a two-echelon model representing the integrated problem of retailer-producer are developed, considering the MLOR rule as a decision variable. Both models are then compared with their respective traditional settings, where the MLOR rule is established *a priori* by the retailer and cannot change. With the goal of establishing relationships between the MLOR rule and three indicators - average freshness, food surplus/waste and profits, several instances are randomly generated varying the type of demand (low to high), variability of demand (low and high), production lot-size, product shelf life and consumer withdrawal policy.

The main contribution of this work is the innovative approach of defining flexible MLOR rules differentiated by shelf life and operational restrictions of the producer. This proposed approach offers clear advantages in comparison with the traditional settings of fixed MLOR rules. In particular, flexible MLOR rules provide substantial reduction in the amount of surplus/waste and higher profits for both producer and retailer-producer supply chain. The most important insight of this research is that relaxing the MLOR rule by 12% leads to a slightly reduction of the average freshness, at most 1%. Even when the MLOR rule can be relaxed in 16%, the average freshness reduction is lower than 1% for some cases. Using a simple rolling horizon approach to model a context with uncertain demand, we validate our deterministic conclusions. In particular we show that even when demand is uncertain, the flexible MLOR rule benefits both producer and retailer in terms of profits and waste/surplus.

We have demonstrated that for a relatively small time horizon (21 periods), it is easy to obtain an improved MLOR rule. Even with the rolling horizon approach, which considers the demand to be uncertain, we have shown an easy way to optimize the MLOR rule. Nevertheless, to optimize the MLOR rule in practice for larger time horizons (e.g., several months) may require too much effort and complexity for an exact solution method or for the approaches considered in this work. Hence, new approaches or solution methods to optimize the MLOR rule in practice are topics for future research. Another possible direction for future research encompasses a broader supply chain setting. In this paper, we only examine the impact of flexible MLOR rules in a setting of single producer-single retailer. It would be interesting to verify if the same insights obtained are extended to a setting with multiple producer-multiple retailers. Also, an approach that combines demand uncertainty with uncertain consumer behavior could extend the insights of this pioneer study, for example by comparing the myopic and strategic behavior of consumers when buying products near the expiration date (e.g., such as in Farshbaf-Geranmayeh and Zaccour [28]).

CRediT authorship contribution statement

Maria João Santos: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft. **Sara Martins:** Conceptualization, Methodology, Software, Writing – review & editing. **Pedro Amorim:** Conceptualization, Methodology, Writing – review & editing, Supervision. **Bernardo Almada-Lobo:** Conceptualization, Writing – review & editing.

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Appendix A. Model for rolling horizon approach

$$\max \sum_{t\in\mathcal{T}}\sum_{a\in\mathcal{A}}\Big(p\cdot R_{a,t}+s\cdot A_{a,t}+p'\cdot(W^L_{a,t}+W^F_{a,t})-u\cdot Y_t-c\cdot q_t -h\cdot(I_{a,t}-R_{a,t}-A_{a,t}) -h'\cdot I'_{a,t}-v\cdot Q_t-f\cdot\beta_t-o\cdot O_t-[d^*_t-(W^L_{a,t}+W^F_{a,t})]\Big) \quad (A.1)$$

(4), (6), (7), (8), (10), (11), (12), (13), (15), (16)

$$\sum_{a\in A} W^L_{a,t} \le d^*_t\cdot\alpha, \quad \forall t\in\mathcal{T} \quad 17a \quad (A.2)$$

$$\sum_{a\in A} W^F_{a,t} \le d^*_t\cdot(1-\alpha), \quad \forall t\in\mathcal{T} \quad 18a \quad (A.3)$$

(19) – (24)

$$Q_t \ge \sum_{a\in A} R_{a,t}, \quad \forall t\in\mathcal{T} \quad 25a \quad (A.4)$$

(26) – (31)

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